

BIROBIDZHAN, Russian Federation

WMO: 317130

Lat: 48.733N

Lon: 132.950E

Elev: 254

StdP: 14.56

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-25.7	-33.8	0.8	-29.1	-29.8	1.0	-24.8	18.6	0.8	15.2	10.0	0.8	270	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	85.9	72.7	82.9	70.9	80.1	69.5	75.1	82.9	73.2	80.3	71.4	77.9	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	122.7	79.7	70.8	114.5	77.5	69.1	107.9	75.6	39.0	82.6	37.1	80.3	35.5	77.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	14.3	12.8		DB	-34.2	90.8	4.8	3.4	-37.7	93.2	-40.4	95.2	-43.1	97.1	-46.6	99.6
				WB	-34.8	77.9	4.8	1.6	-38.2	79.1	-41.0	80.0	-43.7	81.0	-47.2	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	-6.3	2.6	19.7	40.1	54.6	64.7	70.8	67.6	56.2	39.6	16.3	-3.2	(6)
(7)		DBStd	28.39	8.73	8.77	10.41	7.42	6.94	6.00	4.51	5.02	6.78	7.59	11.66	8.79	(7)
(8)		HDD50	7362	1745	1326	939	308	34	0	0	0	20	331	1011	1648	(8)
(9)		HDD65	11184	2210	1746	1404	746	329	79	8	29	271	788	1461	2113	(9)
(10)		CDD50	2035	0	0	0	12	176	441	644	547	207	8	0	0	(10)
(11)		CDD65	381	0	0	0	0	6	70	186	111	8	0	0	0	(11)
(12)		CDH74	2760	0	0	0	9	124	619	1262	677	68	1	0	0	(12)
(13)		CDH80	646	0	0	0	1	18	157	347	117	6	0	0	0	(13)
(14)	Wind	WSAvg	4.1	2.6	3.4	4.6	5.7	5.4	4.1	3.7	3.6	4.0	4.6	4.2	2.8	(14)
(15)	Precipitation	PrecAvg	29.5	0.3	0.4	1.0	1.5	3.0	3.8	6.7	5.8	3.6	1.9	0.8	0.6	(15)
(16)		PrecMax	39.8	0.9	1.9	2.6	3.8	6.7	9.8	14.7	9.2	9.3	5.6	2.8	2.2	(16)
(17)		PrecMin	21.0	0.0	0.0	0.0	0.1	0.5	1.2	1.6	1.1	0.8	0.2	0.0	0.0	(17)
(18)		PrecStd	5.2	0.3	0.5	0.8	1.0	1.7	2.5	3.1	1.8	2.2	1.3	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	33.3	50.9	73.6	82.8	88.9	90.7	86.7	80.1	68.3	47.0	25.8	(19)
(20)			MCWB	20.7	26.8	39.6	53.1	62.9	72.1	75.3	75.3	67.9	54.2	38.5	23.1	(20)
(21)		2%	DB	17.9	27.4	43.6	64.9	77.4	84.2	87.0	83.1	75.4	61.7	41.1	19.6	(21)
(22)			MCWB	15.5	22.6	34.9	49.4	59.4	69.4	74.1	72.6	64.3	49.4	35.1	17.4	(22)
(23)		5%	DB	12.4	22.7	39.4	59.0	73.3	80.8	83.7	80.4	72.1	56.7	36.4	15.3	(23)
(24)			MCWB	10.4	19.2	32.4	46.2	56.9	68.1	72.8	70.7	62.1	45.9	32.1	13.4	(24)
(25)		10%	DB	8.2	18.3	35.7	54.0	69.1	77.2	80.7	77.7	68.7	52.3	32.8	11.3	(25)
(26)			MCWB	6.8	15.6	30.3	43.1	55.6	66.0	71.3	68.8	60.4	44.0	29.6	9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	27.3	40.0	55.2	64.6	75.0	78.2	77.2	69.8	56.2	39.3	23.8	(27)
(28)			MCDWB	23.5	33.3	49.5	70.6	77.5	86.3	86.9	83.8	77.3	64.6	46.1	25.8	(28)
(29)		2%	WB	15.5	22.9	36.0	50.3	61.5	71.6	75.7	74.6	66.8	51.3	35.7	17.7	(29)
(30)			MCDWB	17.8	26.8	42.2	62.7	73.4	81.6	83.8	80.4	73.1	58.1	39.8	19.6	(30)
(31)		5%	WB	10.5	19.3	33.0	47.2	59.1	69.3	73.9	72.5	64.1	47.9	32.8	13.5	(31)
(32)			MCDWB	12.3	22.7	38.4	57.5	70.0	78.5	81.6	78.2	69.3	54.8	35.8	15.3	(32)
(33)		10%	WB	6.8	15.7	30.6	44.1	56.6	67.3	72.3	70.5	61.8	44.9	29.5	9.6	(33)
(34)			MCDWB	8.3	18.3	35.6	52.5	66.4	75.3	79.1	75.7	67.3	51.4	32.9	11.2	(34)
(35)	Mean Daily Temperature Range	MDBR	24.3	26.8	22.3	20.0	20.6	18.4	16.1	16.5	20.8	20.3	18.8	21.6	(35)	
(36)		5% DB	MCDBR	25.0	28.4	22.9	29.3	29.2	25.1	22.2	20.7	26.0	28.2	20.7	21.2	(36)
(37)			MCWBR	22.4	23.9	15.5	16.4	13.6	12.4	10.9	10.7	14.5	16.2	14.8	18.9	(37)
(38)			MCDBR	24.7	27.7	21.5	26.9	25.0	22.4	19.3	17.6	20.7	25.3	19.4	20.3	(38)
(39)		5% WB	MCWBR	22.1	23.3	14.6	15.6	12.8	12.1	10.6	9.8	12.4	15.6	14.4	18.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.322	0.399	0.547	0.573	0.521	0.506	0.446	0.411	0.429	0.354	0.292	(40)	
(41)		taud	2.170	2.042	1.889	1.714	1.744	1.964	2.052	2.223	2.228	2.000	2.085	2.136	(41)	
(42)		Ebn at Noon	241	254	250	223	225	239	241	252	248	210	204	216	(42)	
(43)		Edh at Noon	25	37	51	69	69	56	50	41	37	39	28	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	590	985	1433	1512	1648	1755	1616	1482	1223	832	552	457	(44)	
(45)		RadStd	17	28	108	162	150	247	158	112	119	66	42	26	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page